

Reading Documents Aloud with Small Neural Voices

Abstract. We describe VoxDoc, a document reader that turns PDF and Word files into natural speech on an ordinary laptop. The system runs a small text-to-speech model entirely on the CPU and highlights each word as it is spoken.

1. Introduction

Most people meet long documents at the worst possible moment: on a train, while cooking, or at the end of a tiring day. Listening is often easier than reading, yet commercial screen readers sound robotic and cloud services send private files to someone else's server. A local reader avoids both problems.

Three goals shaped the design of the first version:

- Run on a mid-range laptop without a dedicated graphics card.
- Start speaking within a second of pressing play.
- Highlight the current word so the listener can follow along.

2. Method

The pipeline has four stages: extraction, sentence splitting, synthesis and playback. Extraction pulls raw text out of each page and repairs the damage that typesetting leaves behind, such as words broken across lines and ligature characters like the one in efficient workflow.

Sentence splitting is harder than it looks. Dr. Smith arrived at 3 p.m., e.g. this, The U.S. grew 3.5% in 2024. A naive splitter would cut that line into five pieces. We rely on a statistical parser instead of regular expressions, and we treat every paragraph as a hard boundary so that headings are never glued to the sentence that follows them.

2.1 Synthesis

Each sentence is synthesized with a state-of-the-art model of only eighty-two million parameters. The model returns audio together with the time at which every word starts and ends, which is exactly what the interface needs for high-

lighting. Synthesis runs about three times faster than real time on our test machine, so the listener never waits for the next sentence once playback has started, even when the document is long and the chosen voice is one of the slower ones.

Because a reader that waits for an entire chapter before speaking feels broken, VoxDoc synthesizes each sentence on its own, caches the result under a hash of the text, voice and speed, streams the first clip as soon as it is ready, and keeps generating the following sentences in the background, so that by the time the listener reaches the end of one clip the next one is usually already waiting on disk.

3. Results

We tested the reader on forty documents, including research papers, contracts and novels. Listeners rated the voices as pleasant and the highlighting as accurate. The most common complaint concerned tables and figures, which the reader currently skips. Pages with running headers were handled correctly after we removed lines that repeat on every page.

3.1 Listening tests

Each participant listened to the same three passages with every preset voice and then chose a favorite. Most people preferred the warm American voice for long reading sessions and the British voices for fiction. Several participants asked for a slower default speed when the text was dense or technical, so the calm preset now starts at nine tenths of normal speed.

We also measured how long a listener waits after pressing play. With an empty cache the first sentence arrived in about one second; with a warm cache it arrived almost instantly, because the audio and its word timings are read straight from disk.

4. Conclusion

A small local model is good enough to read everyday documents aloud. Future work will add support for scanned pages through optical character recognition

and will let listeners blend two voices into a new one.